

Roll No:				Name:			
LOURDES MATHA COLLEGE OF SCIENCE AND TECHNOLOGY							
SECOND SEMESTER B.TECH DEGREE FIRST SERIES EXAMINATION, FEB 2025							
PCCST205 DISCRETE MATHEMATICS CS-A, CS-B							
Max Marks: 40						Time: 2 Hours	
Cognitive levels (L): L1- Remember; L2- Understand; L3- Apply; L4- Analyze; L5- Evaluate; L6-Create.							
Part A							
(Answer All questions. Each carries 4 Marks)						Total Marks: 20	
Q. No		Questions	Marks	CO's	BL		
1		Let A=(1,2,3,4,5,6) and B={a,b,c,d,e}.Find the cartesian product of A and B. How many elements are in P(AXB)?	4				
2		If A ={1,2,3,4} give an example of a relation on A which is reflexive and transitive, but not symmetric.	4				
3		a. List the ordered pairs in relations on{1,2,3} corresponding to these matrices.(2 marks) a) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ b) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ b. Let A = {a,b,c}. List the set of all proper subsets and subsets of A.(2 marks)	4				
4		Draw the Hasse Diagram for set {2,4,5,10,12,20,25} on a relation R={(a,b) a divides b}	4				
5		Illustrate the truth table for pV q and p ∧ q. Explain the connectives.	4				
Part B							
Answer any 3 questions. Each carry 10 Marks						Total Marks: 20	
6	a	Find the number of integers between 1 and 1000 inclusive which are divisible by 5, 6 or 8	5				
	b	Let R be a relation on Z by xRy 4 (x-y). Then find all equivalence classes	5				
7	a	If f, g and h are functions on the set of all real numbers. f(x)=x+2, g(x)=x-2, h(x)=3x, then find 1. (f o g) o h 2. h o g o f 3. f o (g o h)	6				
	b	Let m be an integer with m > 1. Show that the relation R ={(a,b) a ≡ b (mod m)} is an equivalence relation on the set of integers.	4				
8	a	Check whether the propositions p ∧ (~ q V p) and p V (q ∧ ~ q) are logically equivalent	4				
	b	Show that ((p→q) ∧ (q→r)) → (p→r) is a tautology	6				